# Investigating Spurious Cue Sensitivity in Semantic Segmentation

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**Abstract.** In this paper, we investigate model reliance on different types of spurious features in the semantic segmentation task. Shortcut learning occurs when models learn to rely on spurious yet easily accessible cues rather than task-relevant features, leading to biased predictions and poor generalization. Although shortcut learning is well studied in image classification, its role and impact in semantic segmentation remain largely unexplored. We investigate this by training several semantic segmentation models on systematically biased datasets in image segmentation, in which we manipulate the color of the object (BCBD), or object size, position, color-patches (ISIC). Our findings demonstrate that both object co-occurrence and color can serve as spurious cues in semantic segmentation models, even though the classification is at the pixel level, highlighting the need to account for shortcut learning in semantic segmentation. The source code and documentation are available at the repository (https://github.com/muhammadahsan/ShortcutsImageSegmentation).

**Keywords:** Shortcut Learning · Semantic Image Segmentation

## 1 Introduction

Shortcuts are spurious predictive cues that improve performance on benchmark datasets but fail under distribution shifts. Shortcut learning occurs when models exploit these non-causal correlations, such as hospital-specific markers, acquisition artifacts, or scanner-specific noise, rather than learning task-relevant representations. [15]. Semantic segmentation assigns a class label to each pixel, thereby facilitating fine-grained scene understanding [12,35], as well as supporting applications such as image-guided interventions and radiological diagnostics [14,31]. Segmentation tasks are now largely led by deep learning architectures such as CNNs and U-Nets [6,13], which consistently deliver top performance

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across multiple imaging domains. Nevertheless, these approaches often demand a substantial amount of precise annotations, making the process expensive and labor-intensive, which is especially an issue in medical domain tasks [24,36]. Consequently, semantic segmentation models are often trained on limited datasets, increasing their susceptibility to shortcut learning, whereby models exploit spurious, non-causal correlations instead of task-relevant features, ultimately degrading generalization [3,18,19,21,22]. Deep learning models remain vulnerable to shortcut learning, despite their success in medical image analysis, relying on spurious correlations rather than task-relevant features.[30,32]. However, compared with image classification, shortcut learning has received considerably less attention in semantic segmentation. Although some works have highlighted potential shortcut mechanisms, such as position-based biases introduced by center-cropped ISIC images [21], a comprehensive understanding of shortcut learning in segmentation models is still lacking. In this study, we fill this gap and we examine shortcut learning within semantic segmentation by constructing controlled shortcut scenarios. We address the following questions: i) Are segmentation models susceptible to shortcut learning? ii) Do different spurious feature types such as object co-occurrence and color impact segmentation? and iii) How different shortcuts affect models' robustness and generalizability?

## 2 Related Work

### 2.1 Deep Learning for Image Segmentation

Deep learning has revolutionized semantic segmentation for both natural and medical images, as extensive surveys indicate. Techniques in this field are categorized into areas such as network architectures (like FCN, U-Net, and UNet++), data augmentation, few-shot semantic segmentation, customized loss functions, weakly supervised approaches, and multi-task learning strategies [1,14,24,31,36]. Models like U-Net and UNet++ leverage encoder–decoder frameworks and integrate skip connections to efficiently combine multi-scale features, resulting in robust performance across various imaging modalities including CT, MRI, and ultrasound [24,36]. Medical image annotation requires expert radiologists or clinicians, making it costly and time-intensive and resulting in limited labeled datasets. Data scarcity and class imbalance remain key challenges, motivating research into weakly, semi, and unsupervised learning, alongside data augmentation and GAN-based synthesis to improve model learning from limited data [31]. Large-scale segmentation foundation models, such as MedSAM, are trained on millions of image–mask pairs to overcome task-specific limitations and enhance generalization across diverse segmentation tasks [14]. However, these advanced models remain vulnerable to biases inherited from their training data and may encode shortcuts if these are not explicitly addressed.

### 2.2 Shortcut Learning in Image Segmentation

Shortcut learning is widely recognized in machine learning, where models exploit spurious correlations between labels and easily accessible features such as tex-

ture, background, or secondary objects [4,17,19,33]. Surveys on spurious correlations ("Clever Hans" behavior) catalog numerous examples and taxonomize mitigation strategies, understanding that shortcuts are especially harmful under distribution shift [17]. In medical AI, shortcut learning can arise from data acquisition protocols, scanner-specific artifacts, demographic imbalances, clinical workflow markers, or treatment devices, producing models that fail when deployed in new settings [18,22,25]. Most prior shortcut work targets classification; segmentation has been comparatively underexplored. Toolkits for fairness and bias mitigation exists but have been evaluated mainly in non-medical domains and rarely in segmentation. [3]. Additionally, techniques have been introduced to identify or investigate shortcut features within a model's internal representations [5,19,25]. By analyzing prediction difficulty and divergence across different network layers, these methods can pinpoint where shortcut features first appear [5,25]. Fairness assessment in ML models requires analyzing performance across various population subgroups to determine if unequal outcomes stem from bias or shortcut learning where models depend on non-representative patterns. Multitask shortcut testing offers a method to directly evaluate whether clinical models use sensitive attributes, such as skin tone, as shortcuts. [7]. However, there is little systematic work that: (i) targets segmentation specifically, (ii) studies multiple shortcut types (e.g., color, position, size, patches) under controlled settings, and (iii) evaluates the relationships between shortcut reliance, limited data availability, and dataset construction, which are key factors influencing medical segmentation models.

### 2.3 Shortcut Learning Mitigation

Feature disentanglement methods mitigate shortcut learning in medical imaging by separating task-relevant features from confounder-related signals. Combined with data-centric rebalancing, these approaches have been shown to reduce shortcut reliance under strong confounding conditions [34]. Intermediate-layer knowledge distillation from specialist teachers trained on reliable subsets guides student models toward robust features, reduces shortcut learning, and improves out-of-distribution performance despite limited bias annotations [5]. Eye-gaze-guided transformers incorporate radiologists' visual attention to steer models toward clinically relevant regions and away from background shortcuts, improving both generalizability and interpretability under limited data [23]. Augmentation strategies based on causal inference, which eliminate misleading correlations between objects' core features and labels that can enhance the robustness of segmentation models when applied across different data acquisition settings [26]. Another weakly supervised approach that operates without model retraining and remains effective even with few or no group annotations. This lightweight framework uses sparse autoencoders to separate spurious features from essential ones, thereby reducing shortcut learning [2]. Methods explicitly targeting confounders or spurious correlations, such as applying causal interventions or leveraging textual guidance in segmentation, are suggested to enhance resilience to domain shifts and minimize dependencies on domain-specific artifacts [10,20].

These findings emphasize that shortcut learning poses a significant barrier to building reliable and trustworthy models for medical imaging.

### 2.4 Scope of This Work

Given the evidence that deep neural networks are susceptible to shortcut learning and fairness problems, this paper contributes a systematic study of shortcut learning in semantic segmentation. Using controlled shortcuts e.g., object color, position, size, and artificial color-patches on medical dataset such as ISIC [11] and non-medical BCBD [27] benchmarks, we investigate how segmentation models exploit shortcuts, how this undermines generalization in data-limited regimes, and how such behavior can be systematically characterized.

## 3 Methodology

In this study, we evaluate how resilient the semantic segmentation model is to misleading or irrelevant correlations in the data. This section outlines the datasets used, describes the models, and explains the metrics for evaluation.

### 3.1 Datasets

To investigate the impact of spurious features on segmentation models, we utilize two datasets: The International Skin Imaging Collaboration dataset (ISIC) [11] and a novel black-cat-brown-dog (BCBD) dataset [27], enabling us to investigate the impact of spurious object co-occurrence, object size, and color features on performance. Figure 1 illustrates the constructed datasets used to analyze shortcut learning in semantic segmentation tasks under different spurious correlation settings. In Fig. 1 (a) BCBD contains independent and identically distributed (IID) and out-of-distribution (OOD) data samples of cats and dogs images along with their corresponding segmentation masks. During dataset construction, object color is intentionally correlated with class labels, creating a spurious shortcut cue. Figure 1 (b) demonstrates lesion-size shortcuts in ISIC skin lesion segmentation. The original lesion and its corresponding mask are resized to approximately 30% of the original lesion size and then repositioned randomly within the image while preserving lesion-mask alignment. The objective is to evaluate whether the segmentation model learns genuine lesion features or instead depends on shortcut cues related to lesion size and spatial position. Figure 1 (c) illustrates patch-induced shortcuts, where artificial colored patches are added to images while maintaining the original lesion masks. Although the lesion structure remains unchanged, the added patches could provide an easier predictive signal for the model to exploit. Consequently, patches that cannot be masked during training appear solely in the input images as potential spurious features and are never treated as target labels. This design allows us to assess the model's susceptibility to shortcut learning while avoiding label leakage in the segmentation masks.

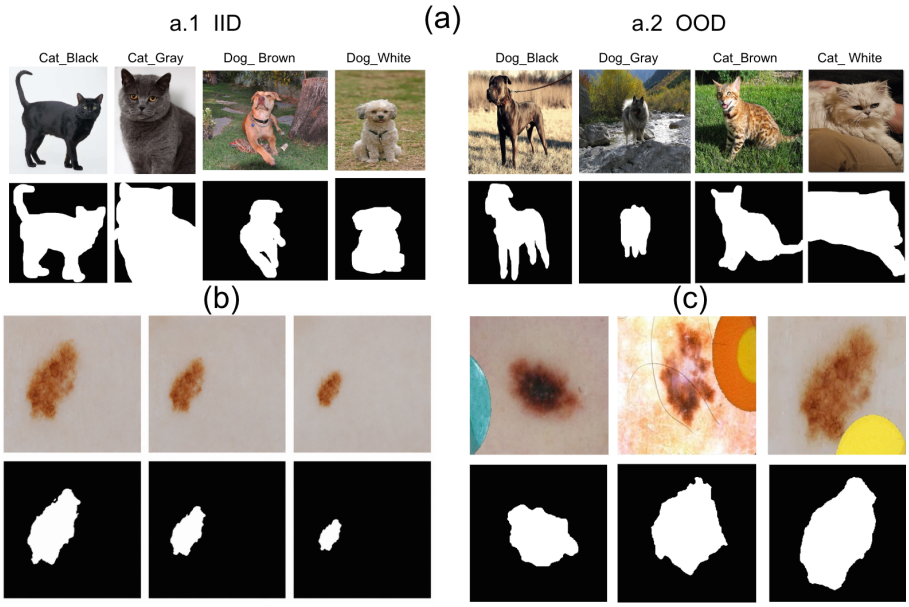



**Fig. 1.** Shortcut-based datasets: (a) IID and OOD samples of BCBD (b) ISIC lesion size and position shortcuts, and (c) ISIC artificial color-patch shortcuts.

**International Skin Imaging Collaboration Dataset (ISIC):** It is widely used for research on skin cancer detection and contains images labeled as benign or malignant [11]. ISIC contains non-clinical artifacts, such as color patches, rulers, and hairs, that can induce spurious correlations. We evaluated the effect of color patches, which appear only in benign training images and in approximately 50% of benign cases. To assess lesion-size bias, lesion size was empirically reduced to approximately 30% of its original size. Reducing the lesion to 50% of its original size produced only marginal changes in model behavior. Further reduction beyond 30% would substantially diminish diagnostically relevant features, as malignant lesions in the ISIC dataset already occupy a small image area, thereby confounding the analysis. To address this, we created a synthetic test set in which color patches were systematically introduced to around 50% of benign & malignant test examples to mitigate their confounding influence. For our analysis, the data is categorized into four groups: BwoP (benign without patch), BwP (benign with patch), MwoP (malignant without patch), and MwP (malignant with patch), where MwP being the minority group. For our experiments, both the training set and the testing set, the proportion of these subgroups in percentage within the class Y (benign, malignant) is defined as Train-set has 6,314 (53%) of BwoP, 5,526 (47%) of BwP, 1,571 (100%) of MwoP, and 0 (0%) of MwP samples, whereas Test-set has 3,158 (53%) of BwoP, 2,763 (47%) of BwP, 821 (50%) of MwoP, and 821 (50%) of MwP samples.

**Black Cats Brown Dogs Dataset (BCBD):** We construct the Black Cats Brown Dogs (BCBD) dataset from the Oxford-IIIT Pet Dataset [27], which provides pixel-level segmentation masks for images of various cat and dog breeds. To obtain color-specific subgroups, we first applied an automated color detection pipeline to the segmented animal regions and subsequently verified and corrected the assignments through manual annotation, ensuring reliable color labels. We treat the 170 black and 43 gray cats, whereas 579 brown and 65 white dogs as training samples. For IID and OOD testing we have 220 and 929 samples respectively. While 589 black dogs and 266 brown cats are used as the OOD testing subgroups where no image in OOD has been used for model training.

### 3.2 Models

We evaluate the widely used standard computer vision architecture including UNet [28], TransUNet [9], and SwinUNet [8]. Specifically, we implemented a UNet model with a ResNet34 [16] encoder backbone, initially pretrained on ImageNet-1K [29], and further fine-tuned it for each task using our datasets. Given the limited training data, we fine-tune the pre-trained UNet models which is a standard transfer learning technique used in low-data training regimes. We used another hybrid model TransUNet that integrates Transformer modules with the U-Net framework for medical image segmentation. Another pure transformer-based model, SwinUNet, is inspired by the U-Net architecture and performed well on medical image segmentation tasks.

### 3.3 Metrics

**Dice Score** is commonly used in image segmentation and relies on the Dice coefficient, which quantifies the overlap between two samples—such as the predicted and ground-truth regions. It is mathematically similar to the F1 score, with values ranging from 0 (no overlap) to 1 (perfect overlap). We used Dice coefficient and IoU as our evaluation metrics. The Dice coefficient (DC) is defined as follows:

$$\mathrm{DC} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} = \frac{2|X \cap Y|}{|X| + |Y|} \tag{1}$$

**The Jaccard Metric,** also known as intersection over union (IoU), measures the similarity between predicted and actual segmentations. It is calculated as the size of the intersection divided by the size of the union of the predicted (X) and ground truth (Y) regions, using true positives (TP), false positives (FP), and false negatives (FN).

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} = \frac{|X \cap Y|}{|X| + |Y| - |X \cap Y|} \tag{2}$$

### 3.4 Evaluation Protocol

We implemented U-Net with a ResNet34 encoder, TransUNet, and SwinUNet, and evaluated them on both IID and OOD test sets described in the Datasets section. To assess robustness, experiments were repeated with three random seeds, producing consistent results. The BCBD and standard ISIC test sets contained no synthetic confounders, whereas a balanced test set for ISIC was created with 50% of samples containing synthetic confounders.

## 4 Experiments and Results

The proposed UNet model is dataset-independent and applicable to multi-class semantic segmentation. Its performance was evaluated on the ISIC and BCBD datasets, with Dice and mIoU results summarized in Table 1. Figure 2 shows that the U-Net model struggles to distinguish Cats from Dogs under OOD conditions, despite accurate segmentation in the IID setting. This suggests reliance on superficial visual cues rather than robust semantic representations.

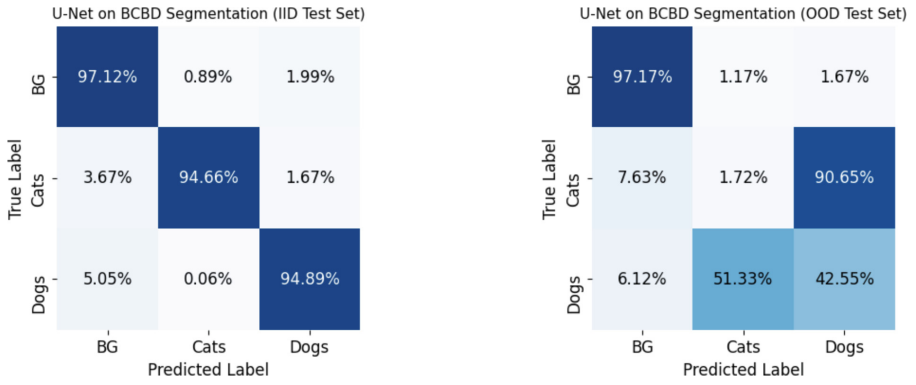



**Fig. 2.** demonstrates that UNet has strong foreground-background learning, but poor semantic discrimination between Cats and Dogs.

The results in Table 1 indicate that the UNet model is relying on the artificial color patches as shortcut features rather than learning robust lesion-specific representations for segmentation. The artificial patches act as spurious cues that are strongly associated with particular labels. When artificial color patches and lesions reduced to 30% of their original size are introduced, the Dice score drops sharply provide strong empirical evidence that the model is learning shortcut correlations induced by color patches and lesion size rather than robust lesion-aware representations. The results in Table 2 demonstrate that U-Net achieves high Dice and mIoU scores when object colors match the training data, with Dice scores above 92% across all groups. However, when object color is changed and used as a shortcut, segmentation accuracy drops sharply especially for

**Table 1.** UNet Semantic segmentation performance on the ISIC dataset under shortcut settings: Color Patches (left) and Lesion Size (right), evaluated using Dice and IoU across Background, Benign, and Malignant classes.

| Color-Patches shortcut test | | | Lesion-size shortcut test | | |
|---|---|---|---|---|---|
| Subgroups | Dice | mIoU | Subgroups | Dice | mIoU |
| Benign_NoPatch | 83.6±0.06 | 71.3±0.0 | Benign_BigCenter | 82.1±0.0 | 70.3±0.0 |
| Malignant_NoPatch | 69.1±0.01 | 53.6±0.01 | Malignant_BigCenter | 69.2±0.0 | 39.7±0.6 |
| **Baselines:** | | | | | |
| Benign_NoPatch | 83.6±0.0 | 71.6±0.0 | Benign_BigCenter | 82.1±0.0 | 70.8±0.0 |
| Benign_Patch | 62.0±0.0 | 46.9±0.0 | Benign_SmallRand | 45.0±0.01 | 39.7±0.6 |
| Malignant_NoPatch | 69.1±0.01 | 53.6±0.01 | Malignant_BigCenter | 69.2±0.0 | 55.8±0.0 |
| Malignant_Patch | 50.0±0.01 | 37.6±0.01 | Malignant_SmallRand | 18.4±0.01 | 09.1±0.01 |

Cat_Brown and Cat_White, where Dice scores fall from over 92% to nearly zero. This significant decline indicates that the model relies on color for the pixel level classification, rather than learning more robust features like shape, texture, or structural representations of the animals.

**Table 2.** UNet Semantic segmentation performance on the BCBD dataset under shortcut settings: on identically distributed (left) and out-of-distribution (right) data, evaluated using Dice and IoU across Background, Cat, and Dog classes.

| Color as shortcut | | | | | |
|---|---|---|---|---|---|
| IID validation set | | | OOD test set | | |
| Subgroups | Dice | mIoU | Subgroups | Dice | mIoU |
| Cat_Black | 96.1 ± 0.02 | 92.7 ± 0.01 | Cat_Brown | 6.2 ± 0.03 | 4.0 ± 0.02 |
| Cat_Gray | 92.3 ± 0.07 | 86.6 ± 0.01 | Cat_White | 1.0 ± 0.01 | 0.00 ± 0.00 |
| Dog_Brown | 94.9 ± 0.03 | 90.3 ± 0.11 | Dog_Black | 48.8 ± 0.05 | 36.6 ± 0.05 |
| Dog_White | 95.4 ± 0.06 | 91.2 ± 0.02 | Dog_Gray | 50.4 ± 0.10 | 34.4 ± 0.11 |

### 4.1 Ablation Study

We investigated the effects of shortcut learning through ablation studies on the BCBD [27] and ISIC [11] datasets using UNet, TransUNet, and SwinUNet. These experiments were designed to systematically assess the influence of shortcut learning across both convolutional and transformer-based segmentation models, enabling a comprehensive evaluation of model performance and generalization.

**TransUNet** is a hybrid CNN–Transformer architecture for medical image segmentation [9]. It uses Transformer-encoded CNN patch tokens to capture global context, while the U-Net decoder fuses them with high-resolution CNN features for precise localization.

**SwinUNet** is a Transformer-based model inspired by the U-Net architecture, designed for medical image segmentation [8]. Images are divided into patches and processed through a U-shaped encoder-decoder framework utilizing skip connections to capture both local and global semantic information. The encoder leverages a hierarchical Swin Transformer with shifted windows to extract contextual features from the input data. Figure 3 presents in-distribution (ID) and out-of-distribution (OOD) segmentation results of SwinUNet and TransUNet on the BCBD dataset, where object color acts as a shortcut feature. Left and right plots correspond to ID and OOD groups, respectively. Performance is evaluated using the Dice coefficient and IoU across pet species and color combinations. Both models perform well on segmentation tasks when the color distribution is similar to the training data. TransUNet's global attention mechanism enhances its ability to capture contextual information. However, both models show a strong dependence on object color instead of accurately learning true semantic features. The Fig. 4 presents a comparison of SwinUNet and TransUNet on the ISIC skin lesion segmentation dataset, examining two shortcut manipulations: 1. Color-patch shortcut (left panels), 2. Lesion-size shortcut (right panels). The findings suggest that both models often rely on spurious features rather than core features like lesion shape and boundaries for segmentation.

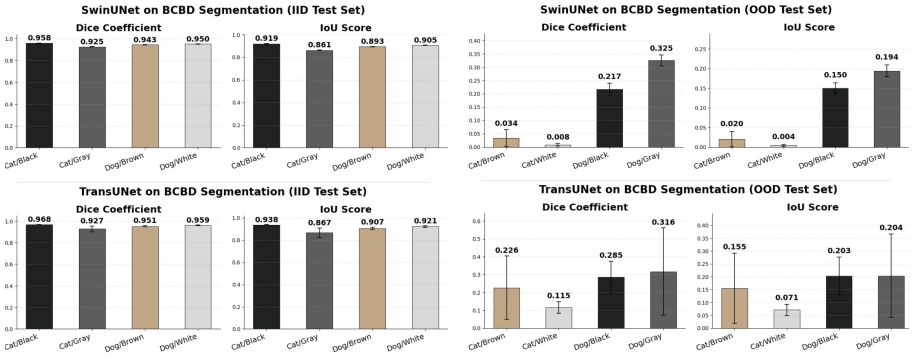



**Fig. 3.** An ablation study on the impact of shortcut learning on SwinUNet and TransUNet performance on the BCBD dataset, with object color as a shortcut feature.

## 5 Discussion and Conclusion

This study examined whether semantic segmentation models can rely on spurious visual cues when such cues are correlated with target labels during training.

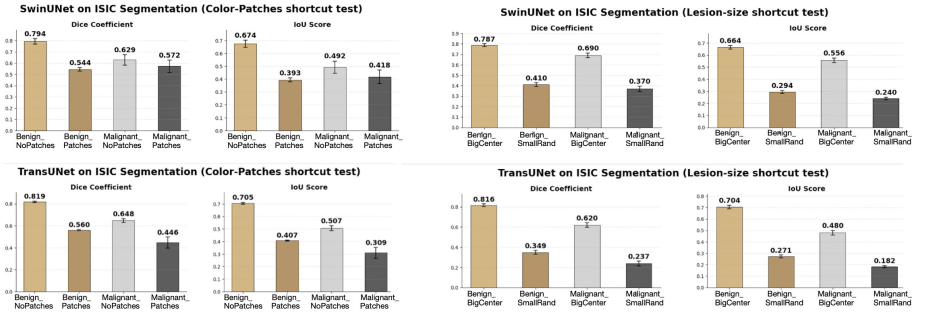



**Fig. 4.** Ablation study on the impact of shortcut learning on SwinUNet and TransUNet performance on the ISIC dataset, with color patches and lesion size as shortcut features.

Using controlled shortcut settings in the ISIC and BCBD datasets, we observed that segmentation performance varied substantially across subgroups when color patches, lesion size/position changes, or object color were introduced as shortcut cues. BCBD dataset, while useful for benchmarking a novel methodology, provides additional insight beyond confirming that segmentation models can exploit intentionally engineered color shortcuts as well as rely on naturally occurring object colors of cats & dogs; it's color-dependent bias. These results suggest that the segmentation models did not perform equally across all subgroup conditions and may have relied, at least in part, on shortcut features that were easier to learn than more generalizable object or lesion representations.

Overall, the findings highlight that semantic segmentation models can be vulnerable to dataset-specific correlations, even when the evaluation target is pixel-level prediction rather than image-level classification. This has practical relevance for medical imaging, where limited data, acquisition artifacts, clinical markers, and subgroup imbalance may introduce shortcuts that are difficult to detect from aggregate performance alone. Subgroup-level evaluation is therefore important when assessing segmentation robustness.

This work has few limitations. First, some shortcut conditions were synthetically created, which allows controlled analysis but may not fully capture the complexity of real-world artifacts. Second, the study focused mainly on performance differences across predefined subgroups and did not include interpretability analyses to directly confirm which image regions the model used for prediction.

Future work could extend this analysis to additional architectures, datasets, and naturally occurring shortcut cues. Systematic evaluation across varying OOD drift levels could better quantify shortcut learning and robustness. Combining subgroup analysis with explainability or counterfactual methods may further reveal reliance on spurious features, enabling controlled evaluation of mitigation strategies such as balanced datasets, targeted augmentation, shortcut-aware validation, and feature disentanglement.

In conclusion, this study demonstrates that shortcut learning can affect semantic segmentation under controlled spurious-correlation settings. The find-

ings highlight the importance of evaluating models beyond aggregate Dice or IoU scores by assessing performance across subgroups with absent, altered, or reversed shortcut cues. Such evaluation revealing hidden failure modes and supporting the development of more robust segmentation systems.

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